

Assignment - 2

Online Multiplayer Gaming

→ Use of Parallel Computing:

- Game servers and client devices (PCs, consoles) use parallel computing to handle multiple tasks at the same time.
- Rendering complex 3D graphics using multi-core GPUs and CPUs.
- Simultaneously managing game physics, animations, AI behaviors, and player input.
- Processing thousands of concurrent game events efficiently.

→ Use of Networked Systems:

- These games rely on networked systems to:
- Enable real-time communication between players across the globe.
- Sync game states between clients and servers to maintain a consistent experience.
- Allow cloud-based matchmaking, player authentication, and game state storage.

→ Why these Technologies are Important:

Performance & Real-Time Interaction:

without parallel computing, the heavy computation load (graphics, logic, AI) would lead to lag and reduced frame rates. Without networked system, multiplayer interaction wouldn't be possible.

- Scalability:

Games can support millions of players simultaneously across different regions.

- Reliability & Speed:

Parallel processing helps in faster load times and seamless transitions; networking ensures timely synchronization and event delivery with minimal latency.